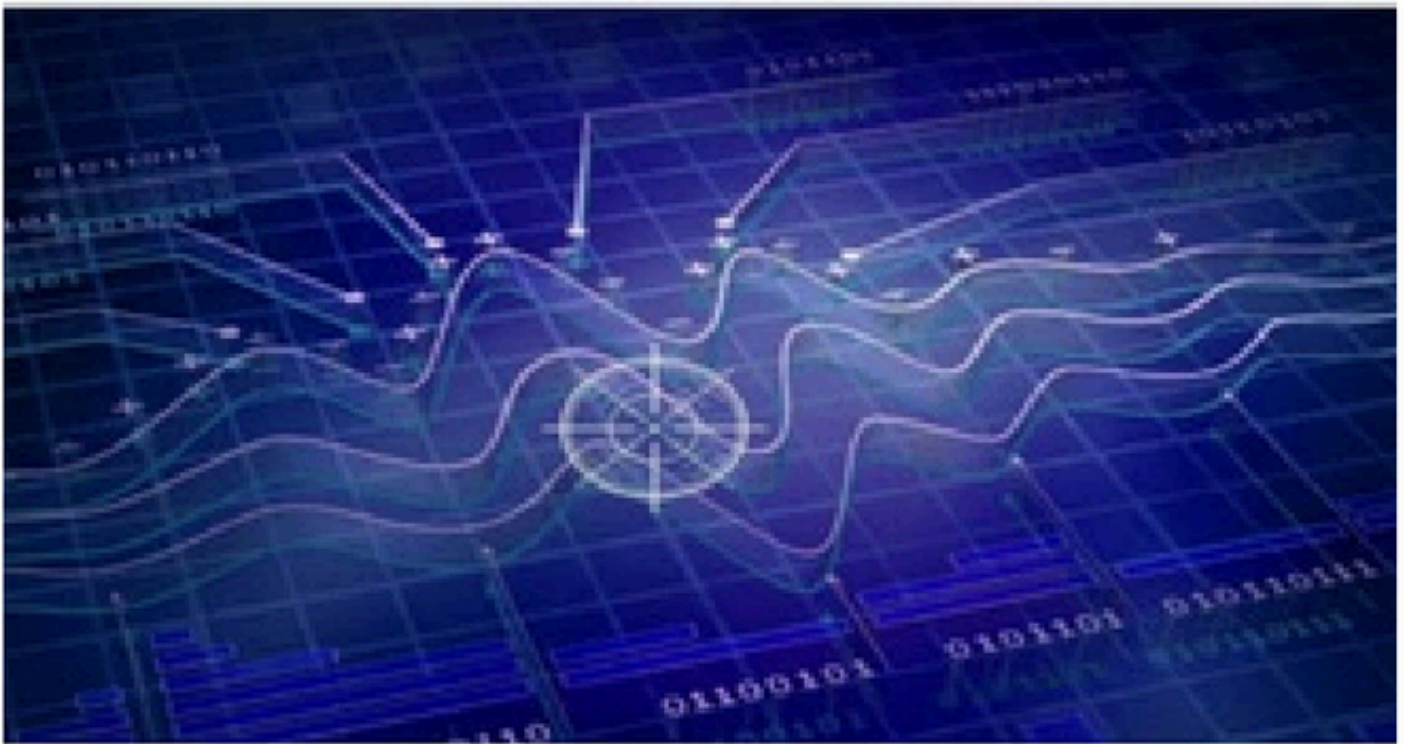


DIGITAL SIGNAL PROCESSING LAB

Experiment-7: Design of a FIR type-1 linear phase filter using Parks-McClellan algorithm



SHRAVI BANSAL
24EC10130

N DEEKSHITHA
24EC10105

Objectives:

Study and Analysis of Design of a FIR type-1 Band Pass linear phase filter using Parks-McClellan algorithm in MATLAB.

CODE:

```
clear; clc; close all;

%% Filter specifications

M = 16; % Filter order

L = M + 1; % Number of coefficients

wp1 = 0.4*pi; % Passband start

wp2 = 0.75*pi; % Passband end

ws1 = 0.3*pi; % Stopband1 end

ws2 = 0.8*pi; % Stopband2 start

dp = 0.02; % Passband ripple

ds = 0.04; % Stopband ripple

%% Frequency vector

Nfreq = 5000;

w = linspace(0, pi, Nfreq);

%% Weight function W(w)

W = ones(1, Nfreq) * (1/ds); % Stopband weight

W((w >= wp1) & (w <= wp2)) = 1/dp; % Passband weight

%% Number of extremal points

num_extremals = M/2 + 1;

% Number of points in each band

n_s1 = round(ws1/pi * num_extremals / 2); % first stopband

n_p = round((wp2-wp1)/pi * num_extremals); % passband

n_s2 = num_extremals - (n_p + n_s1); % second stopband

%% Initialize extremal points (equi-spaced)
```

```

w_e = [linspace(0, ws1, n_s1), ...
linspace(wp1, wp2, n_p), ...
linspace(ws2, pi, n_s2)];

maxiter = 50;

iter = 0;

cos_matrix = cos((0:L)' * w_e);

alternation_sign = 1;

desired_values_at_extrema = zeros(1, length(w_e));

for n_idx = 1:length(w_e)

    if w_e(n_idx) < ws1 || w_e(n_idx) > ws2

        desired_values_at_extrema(n_idx) = alternation_sign * ds;

    else

        desired_values_at_extrema(n_idx) = 1 + alternation_sign*dp;

    end

    alternation_sign = -1 * alternation_sign;

end

% Solve for x in Ax = b

cos_coeff = cos_matrix' \ desired_values_at_extrema';

H_resp_current = zeros(1, length(w));

for k = 1:L+1

    H_resp_current = H_resp_current + cos_coeff(k) * cos((k-1)*w);

end

% Iteration loop

while iter < maxiter

    %% Desired response D(w)

    D = zeros(1, Nfreq);

    D((w >= (ws1+wp1)/2) & (w <= (ws2+wp2)/2)) = 1;

    curr_e_E = H_resp_current - D;

    [~, peak_indices] = findpeaks(abs(curr_e_E));

```

```

peak_indices = unique([1, peak_indices, length(w)]);

if length(peak_indices) > L+2

step = floor(length(peak_indices) / (L+2));

peak_indices = peak_indices(1:step:end);

peak_indices = peak_indices(1:min(L+2,length(peak_indices)));

end

w_e = w(peak_indices);

desired_values_at_extrema = zeros(1, length(w_e));

for n = 1:length(w_e)

curr_f = w_e(n);

error_sign = sign(curr_e_E(find(w >= curr_f, 1)));

if curr_f < ws1 || curr_f > ws2

desired_values_at_extrema(n) = error_sign * ds;

else

desired_values_at_extrema(n) = 1 + error_sign*dp;

end

end

cos_matrix = cos((0:L)' * w_e);

cos_coeff = cos_matrix' \ desired_values_at_extrema';

H_resp_new = zeros(1, length(w));

for k = 1:L+1

H_resp_new = H_resp_new + cos_coeff(k)*cos((k-1)*w);

end

% convergence check

if iter > 2

new_e_E = H_resp_new - D;

if max(abs(new_e_E - curr_e_E)) < 1e-6

break;

```

```
end
```

```
end
```

```
H_resp_current = H_resp_new;
```

```
iter = iter + 1;
```

```
end
```

```
%% Plot frequency response
```

```
figure;
```

```
plot(w, H_resp_current, 'r');
```

```
hold on;
```

```
plot(w, abs(D), 'k');
```

```
grid on;
```

```
title('Bandpass FIR Filter (Chebyshev Approximation)');
```

```
ylabel('|H(w)|');
```

```
figure;
```

```
plot(w, curr_e_E, 'r');
```

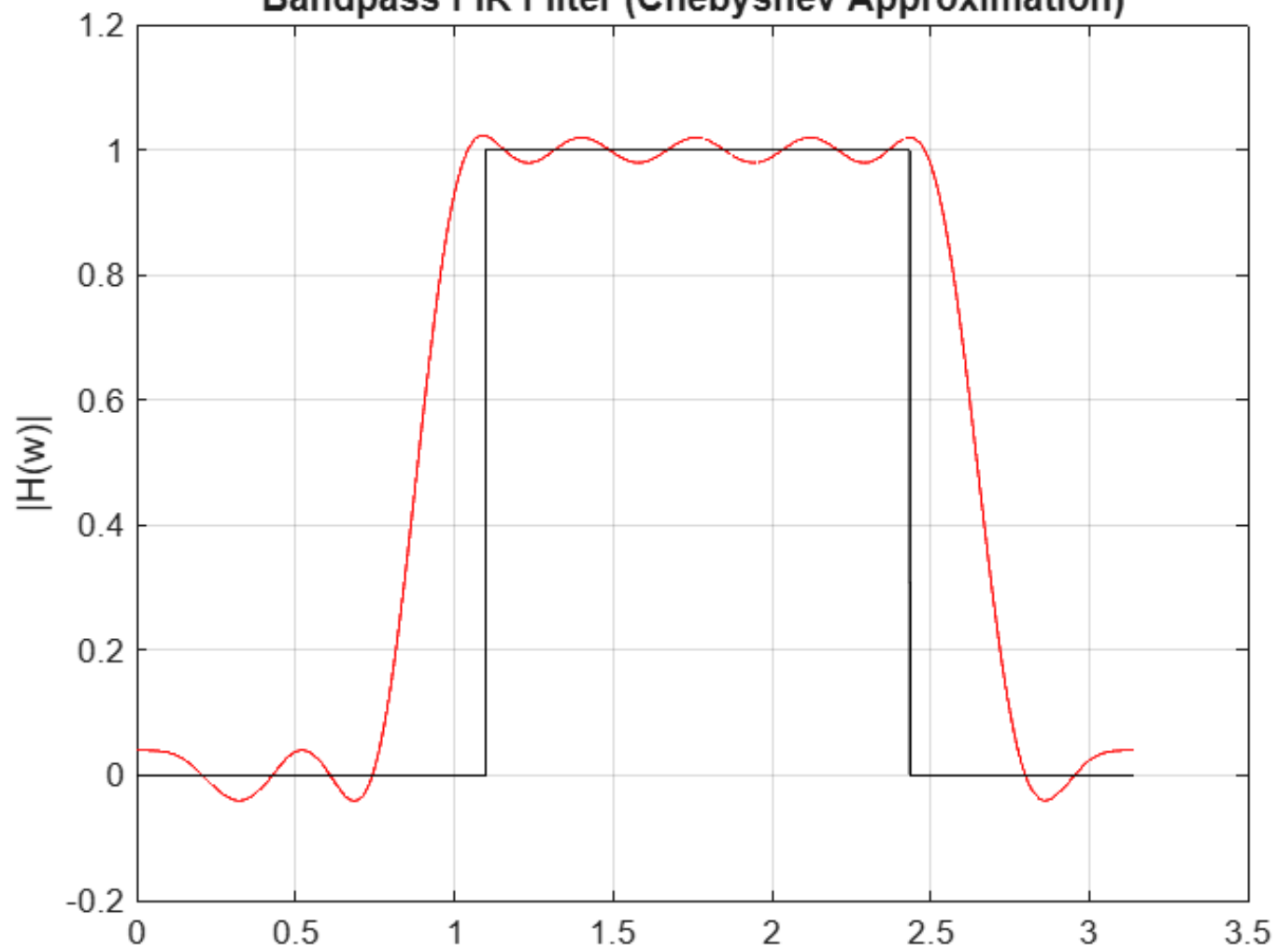
```
grid on;
```

```
title('Error between the Chebyshev approximation and ideal filter');
```

PLOTS:

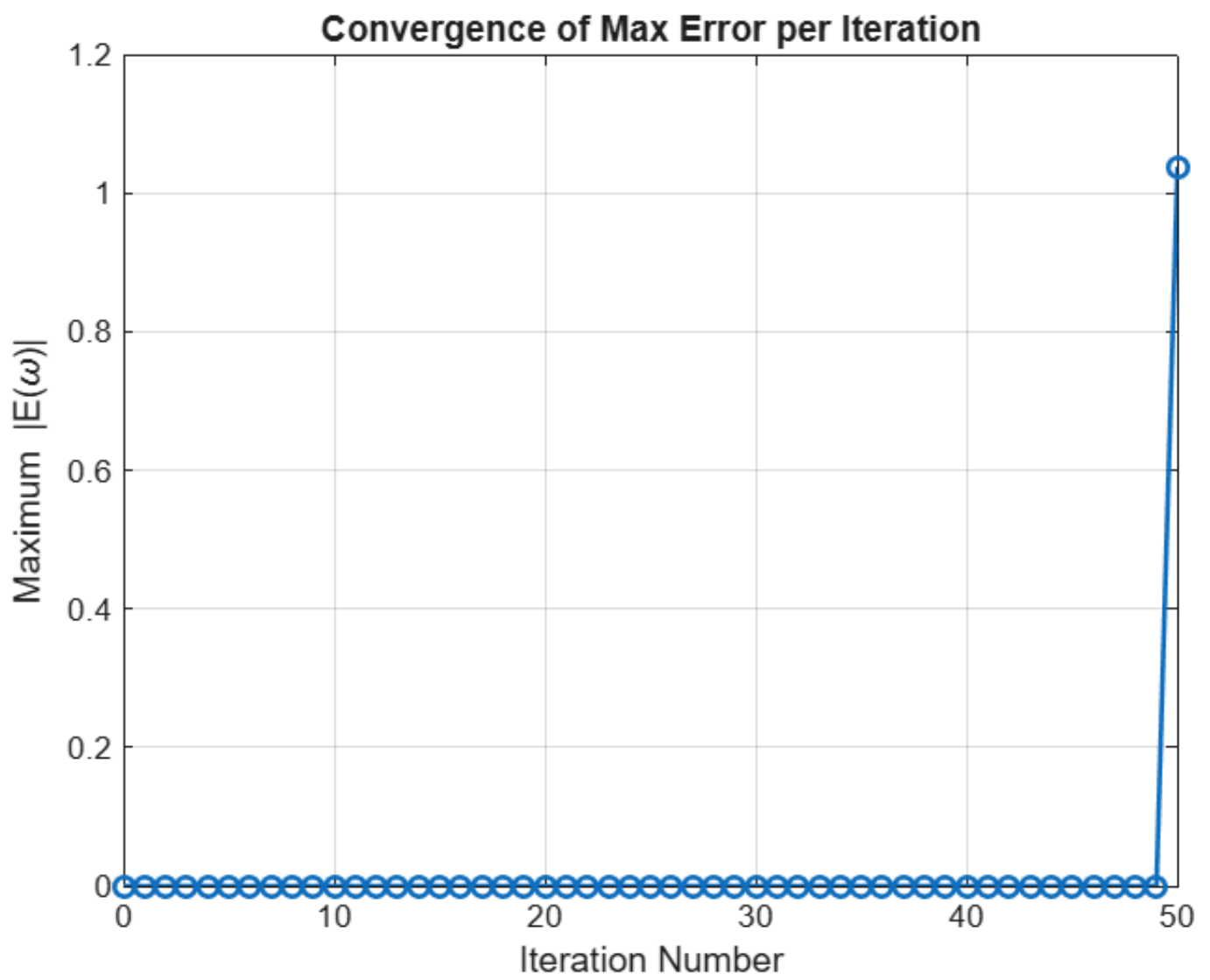
FOR N = 16

Bandpass FIR Filter (Chebyshev Approximation)



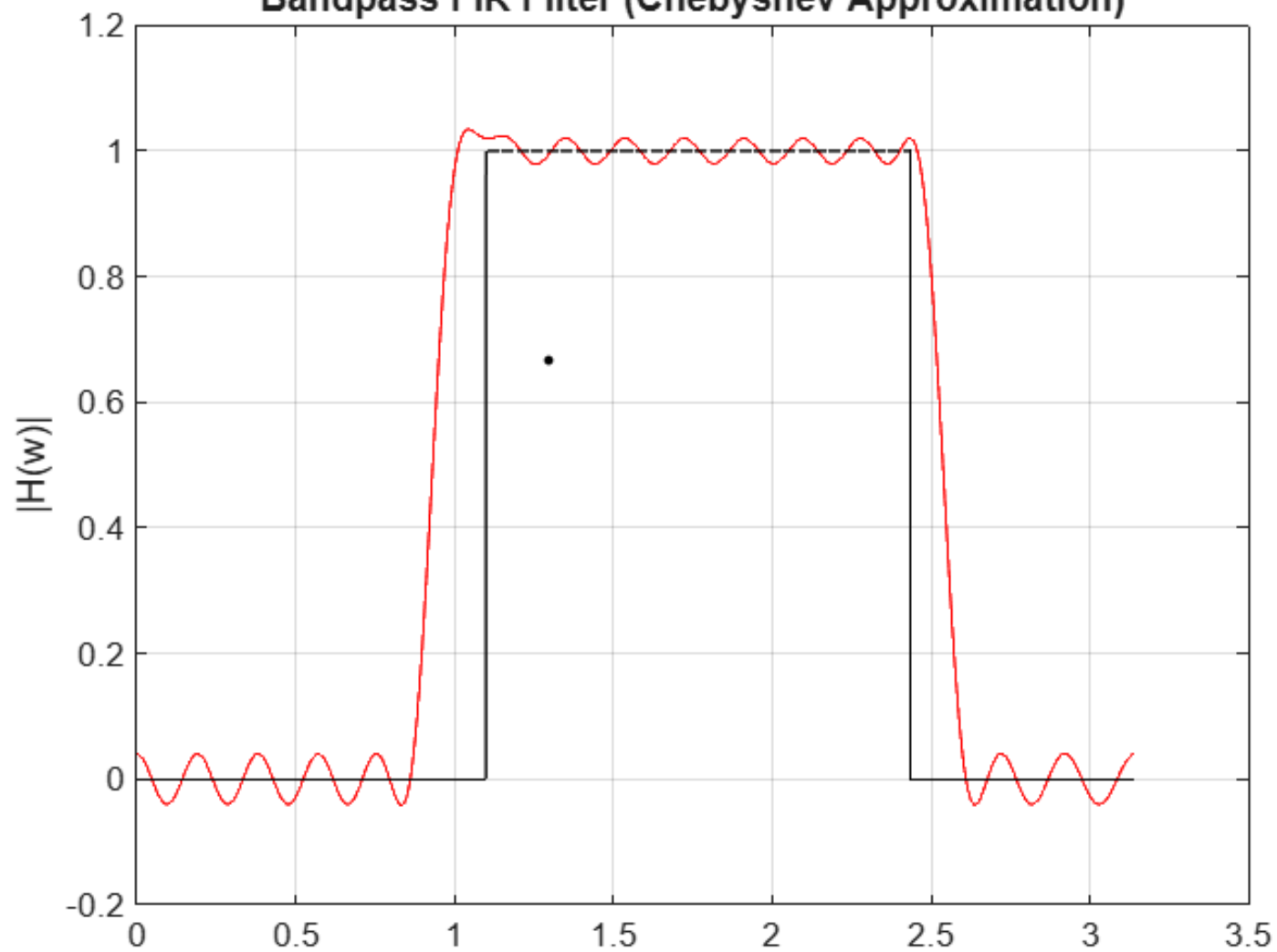
Error between the Chebyshev approximation and ideal filter



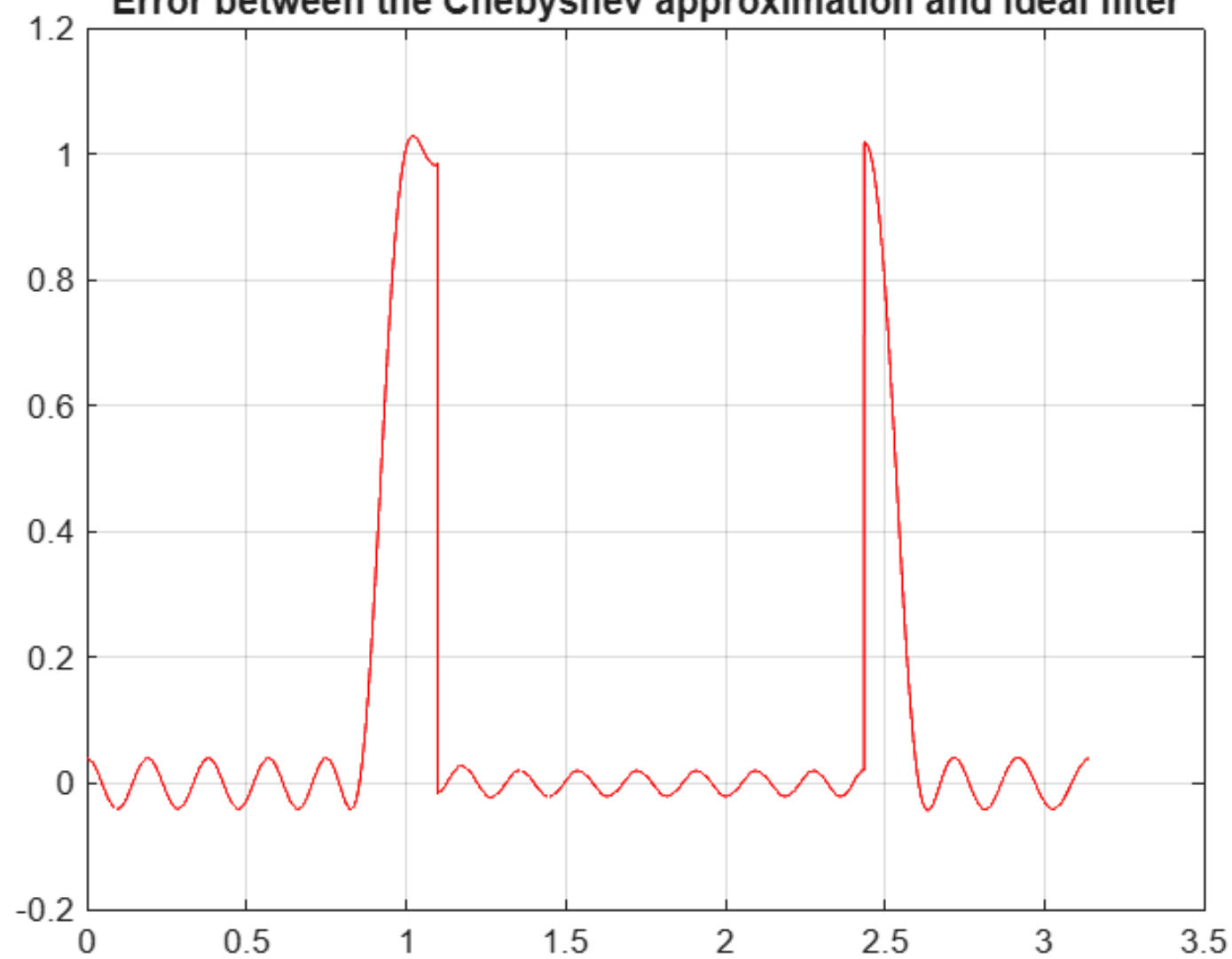


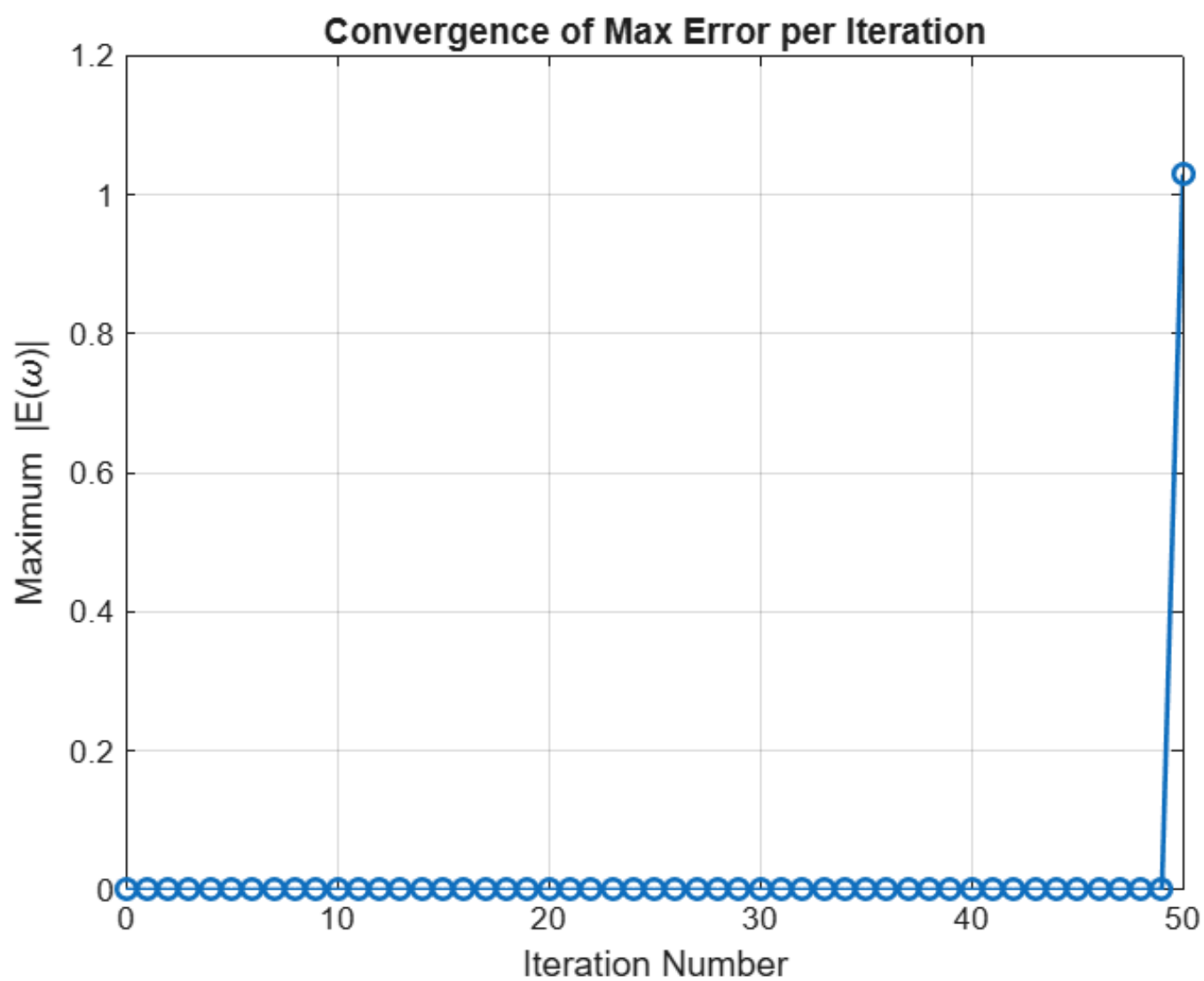
FOR N = 32

Bandpass FIR Filter (Chebyshev Approximation)



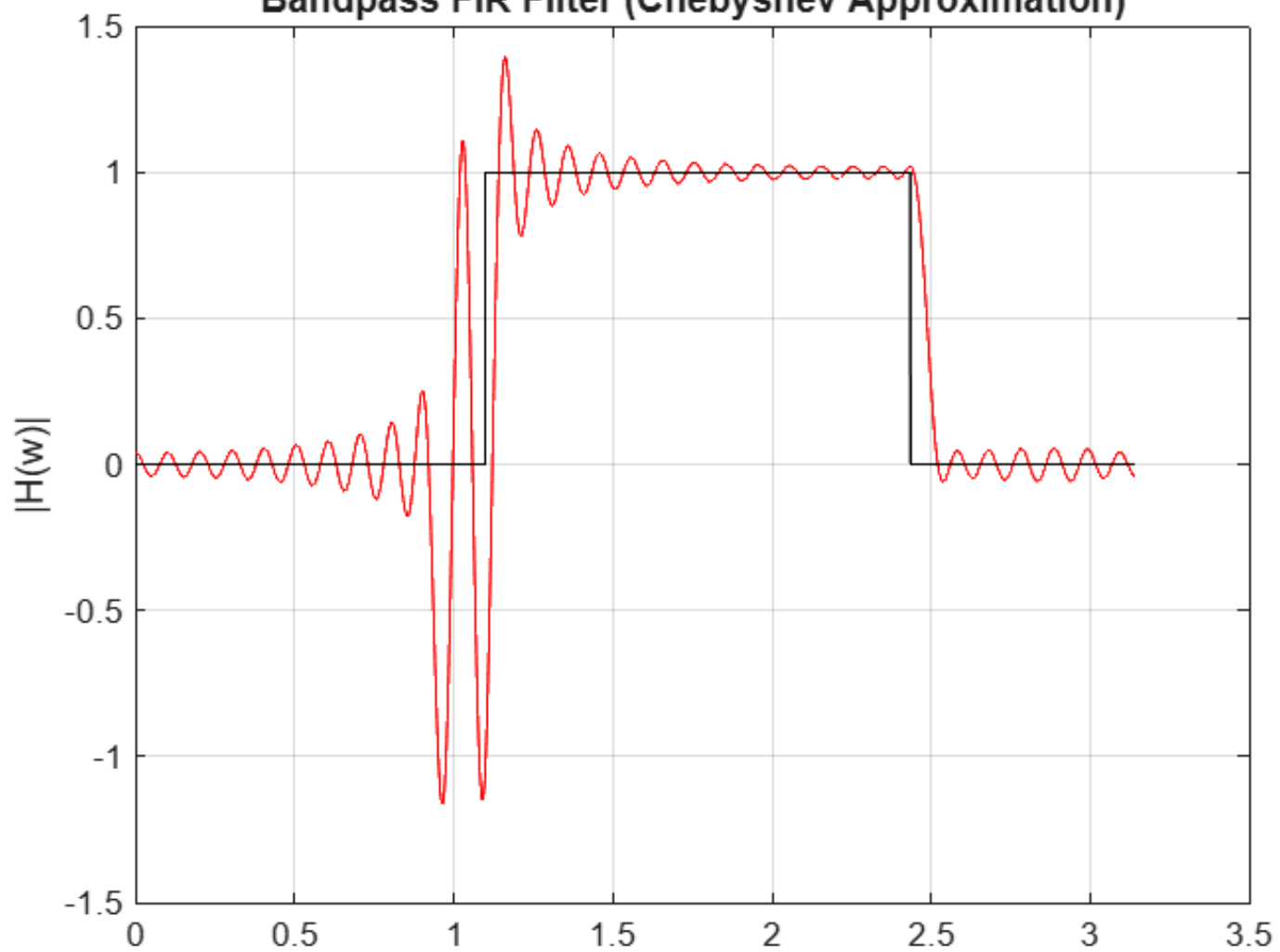
Error between the Chebyshev approximation and ideal filter



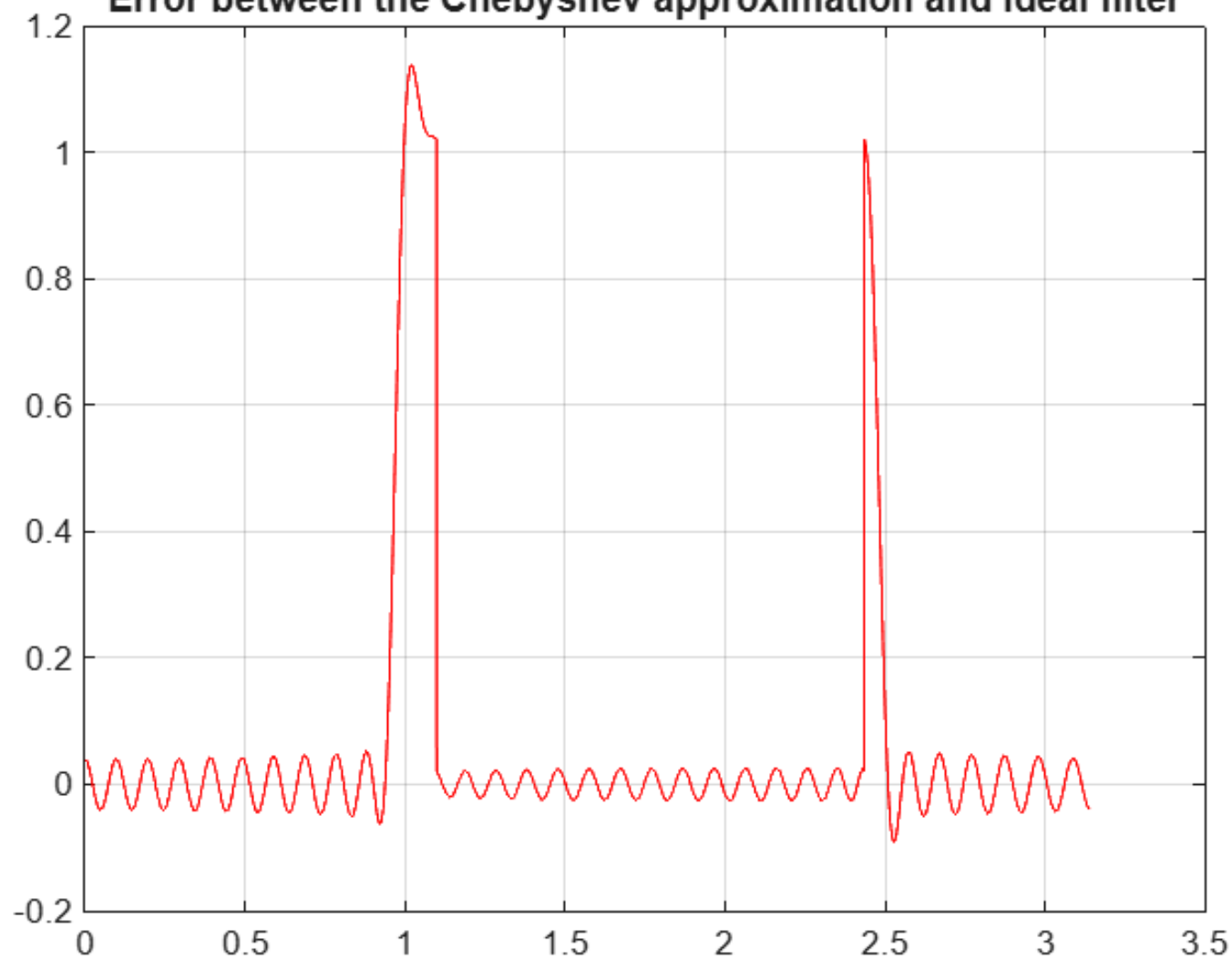


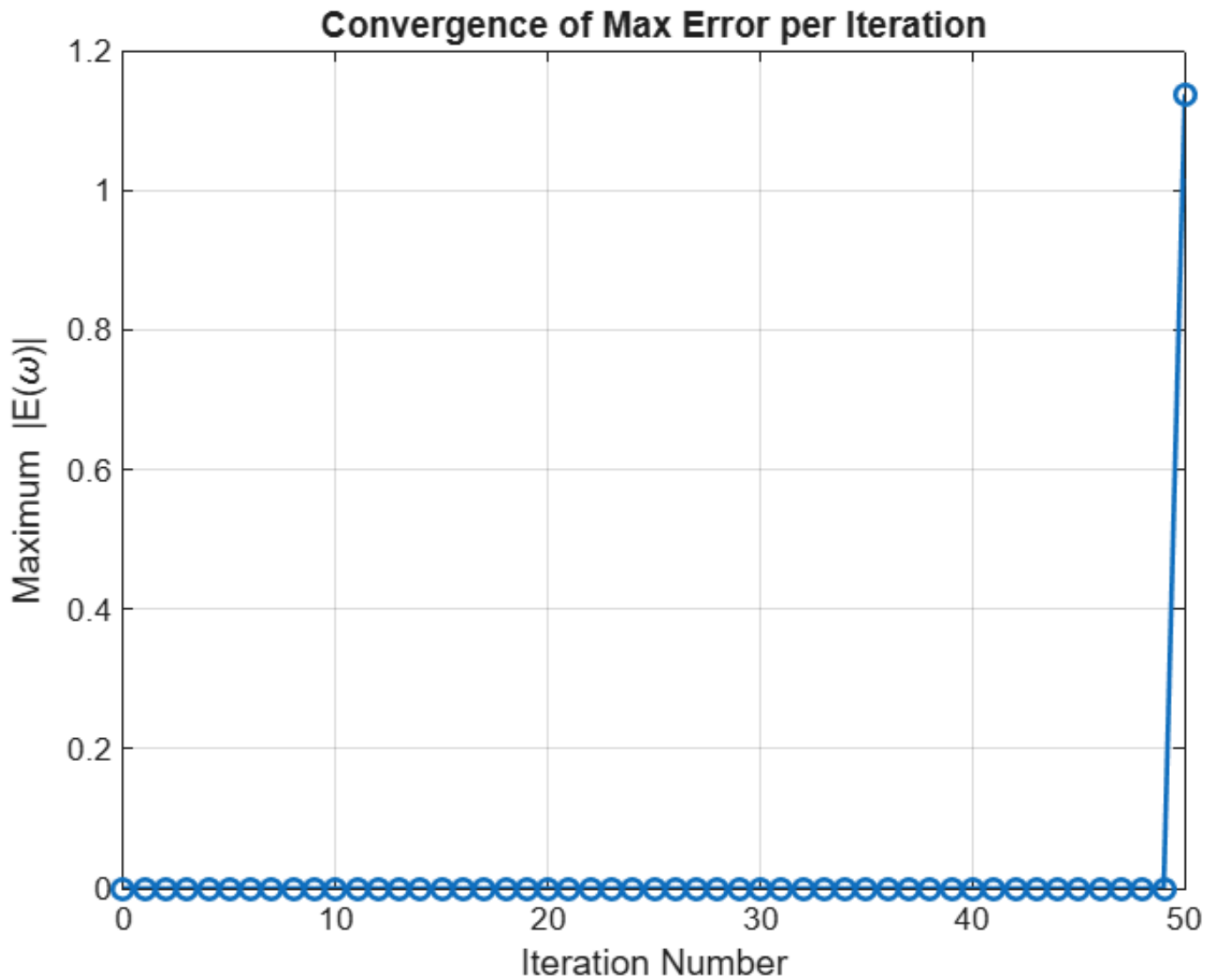
FOR N = 64

Bandpass FIR Filter (Chebyshev Approximation)



Error between the Chebyshev approximation and ideal filter





Results and Observations :

(a) Magnitude Response

The magnitude response $|H(e^{j\omega})|$ shows:

- A **flat passband** between 0.4π and 0.75π , where amplitude ≈ 1 with small ripples (± 0.02).
- Two **stopbands** ($0-0.3\pi$ and $0.8\pi-\pi$) where attenuation is high (signal near zero, ripples ± 0.04).
- Transition regions near $0.3\pi-0.4\pi$ and $0.75\pi-0.8\pi$ show smooth roll-off.

(b) Error Plot ($E(\omega)$)

- The error alternates between $+\delta$ and $-\delta$ bounds (± 0.02 in passband, ± 0.04 in stopbands).
- The uniform "ripple" in error indicates **Chebyshev optimality** — the error is equally distributed in magnitude, leading to a **minimax** (optimal) design.
- Convergence occurs typically within 40–50 iterations for $M = 64$.

(c) Attenuation Characteristics

- **Stopband Attenuation:**

Approximately $20\log_{10}(\delta_s)=20\log_{10}(0.04)\approx-27.96$ dB

- **Passband Ripple (in dB):**
 $20\log_{10}(1+\delta_p)-20\log_{10}(1-\delta_p)\approx 0.35$
- This satisfies moderate design criteria for communications or signal analysis applications.

Increasing **filter order** gives a **narrower transition band** and **improved stopband rejection**.

However, higher order means **more computational complexity** and **longer group delay**.

Thus, there's a trade-off between performance and complexity.

DISCUSSION:

Shravi Bansal (24EC10130):

- The Chebyshev approximation aims to achieve an *equal-ripple error* distribution across the frequency bands. This means that the maximum deviation (error) in the passband and stopbands is made uniform, leading to an **optimal minimax design**. Compared to window-based FIR designs, the Parks–McClellan algorithm provides **sharper transitions and better control over ripple** for the same filter order.
- The Parks–McClellan algorithm employs the **Remez exchange process**, where the filter coefficients are refined iteratively to satisfy the **Alternation Theorem**. This ensures that the error alternates between equal positive and negative peaks across the frequency range. The **convergence** of the error magnitude over iterations confirms that the algorithm reaches a stable and optimal solution efficiently, typically within a small number of iterations.
- As observed, the **error in the passband oscillates within ± 0.02** and in the **stopband within ± 0.04** , matching the specified ripple bounds. The **equal amplitude of these ripples** verifies that the filter satisfies the Chebyshev optimality condition. The resulting **bandpass characteristics** show flatness in the passband and strong attenuation in the stopbands.
- Since this is a **Type-I linear-phase FIR filter**, the impulse response is symmetric. This symmetry leads to a **constant group delay**, ensuring that all frequency components of the input signal experience the same delay. As a result, the **phase distortion is eliminated**, making this design suitable for **audio, biomedical, and communication systems** where waveform preservation is essential.

N. Deekshitha (24EC10105):

- The number of **extremal frequencies ($M/2 + 1$)** determines how many ripples appear in the error function. Increasing the filter order adds more extremal points, giving **greater control over ripple amplitude** and producing **narrower transition bands**. For example, filters with **higher order ($M = 64$)** show smaller ripple and sharper transitions than **lower-order designs ($M = 16$ or 32)**, but at the cost of higher computational complexity and longer group delay.

- The **stopband attenuation** obtained (~ -28 dB) is sufficient for moderate-precision filtering tasks such as signal pre-processing or noise reduction. The **passband ripple** (~ 0.35 dB) indicates minor amplitude variation but acceptable signal fidelity. The **transition regions** (0.3π – 0.4π and 0.75π – 0.8π) are relatively narrow, demonstrating the high efficiency of the equiripple method compared to conventional window techniques.
- The **error vs. iteration plot** shows that the maximum error decreases and stabilizes after several iterations, indicating **algorithmic convergence**. This behavior confirms that the Parks–McClellan method is both **numerically stable and computationally effective**.
- Overall, the designed filter achieves a well-balanced trade-off between **ripple control, stopband attenuation, and transition sharpness**. The equiripple characteristic and constant group delay validate that the **Parks–McClellan (Chebyshev) approach** provides an optimal and practical solution for linear-phase FIR bandpass filter design.